Speech and Language Processing (3rd ed. draft)

Dan Jurafsky and James H. Martin

Here's our August 19, 2026 release!

- This release finally has a chapter 1! Much LLM material from the former chapter 7 was decanted here, plus lots of new material to make this a fun intro chapter for students as well as other newcomers to the field.
- The former chapter 8 (Transformers) was merged with the remainder of former chapter 7 (=pretraining and decoding/sampling) to create a new single chapter that covers transformers, pretraining, and decoding.
- There is a large stub (about half) of an interpretability chapter 10, more to come on the next edition.
- We used Claude Opus 5 to suggest more exercises for various chapters, and also to do a pass over the first 8 chapters to point out any bugs it could find. It found a lot. The result is some more exercises in this release, and also hopefully fewer typos and subtle notation inconsistencies to plague you the readers!

Individual chapters are below. **There are now updated slides for the intro and transformer/pretraining chapters!!**

Here is a single pdf of Aug 19, 2026 book!

1. **Feel free to use the draft chapters and slides in your classes**, print it out, whatever, the resulting feedback we get from you makes the book better!
2. **Typos and comments** are very welcome (just email slp3edbugs@gmail.com and let us know the date on the draft)! (Don't bother reporting missing refs due to cross-chapter cross-reference problems in the indvidual chapter pdfs, those are fixed in the full book draft)
3. **Gratitude!** We've put up a list here of the wonderful people who have sent so many fantastic suggestions and bug-fixes for improving the book. We are really grateful to all of you for your help, the book would not be possible without you!
4. **How to cite the book**:

   Daniel Jurafsky and James H. Martin. 2026. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition. Online manuscript released August 19, 2026. https://web.stanford.edu/~jurafsky/slp3.

5. A **bib entry** for the book is **here**.

```
@Book{jm3,
  author =       "Daniel Jurafsky and James H. Martin",
  title =        "Speech and Language Processing: An Introduction to Natural Language Processing,
                  Computational Linguistics, and Speech Recognition,
                    with Language Models",
  year =         "2026",
  url = {https://web.stanford.edu/~jurafsky/slp3/},
  note = "Online manuscript released August 19, 2026",
  edition =         "3rd",
  }
```

6. **When** will the book be finished? Don't ask.
7. If you need the previous Jan 2026 draft chapters, they are here; If you need the previous Aug 2025 draft chapters, they are here;

Volume I: Large Language Models

	Chapter	Slides
1:	Introduction	1: [pptx] [pdf]
2:	Words and Tokens	2: Words and Tokens [pptx] [pdf] 2: Edit Distance [pptx] [pdf]
3:	N-gram Language Models	3: [pptx] [pdf]
4:	Logistic Regression and Text Classification	4: [pptx] [pdf]
5:	Embeddings	5: [pptx] [pdf]
6:	Neural Networks	6: [pptx] [pdf]
7:	Transformers and Pretraining	7: [pptx] [pdf]
8:	Post-training	

Volume II: Advanced LLM Topics and Tools

	Chapter	Slides
9:	Masked Language Models	9: [pptx] [pdf]
10:	Interpretability	
11:	Information Retrieval and RAG	11: [pptx] [pdf]
12:	Agents [not written yet]	
13:	Machine Translation	
14:	RNNs and LSTMs	13: [pptx] [pdf]
15:	Phonetics and Speech Feature Extraction	
16:	Automatic Speech Recognition	
17:	Text-to-Speech	

Volume III: Annotating Linguistic Structure

	Chapter	Slides
18:	Sequence Labeling for Parts of Speech and Named Entities	18: (Intro only) [pptx] [pdf]
19:	Context-Free Grammars and Constituency Parsing	
20:	Dependency Parsing	
21:	Information Extraction: Relations, Events, and Time	
22:	Semantic Role Labeling and Argument Structure	
23:	Lexicons for Sentiment, Affect, and Connotation	
24:	Coreference Resolution and Entity Linking	
25:	Discourse Coherence	
26:	Conversation and its Structure	

Appendix (will be just on the web)

A:	Hidden Markov Models	
B:	Naive Bayes Classification	B: [pptx] [pdf]
C:	Kneser-Ney Smoothing	
D:	Spelling Correction and the Noisy Channel	
E:	Statistical Constituency Parsing	
F:	Context-Free Grammars	
G:	Combinatory Categorial Grammar	
H:	Logical Representations of Sentence Meaning	
I:	Word Senses and WordNet	
J:	PPMI	
K:	Frame-based Dialogue Systems	